

Report-Space Mechanisms for Informative Transcripts

Turning eigengrades from an estimator into a (money-free) mechanism

(co-mathematician working paper)

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Abstract

Working draft. Gans and Kominers (2026) show that a scalar grade cannot separate student ability a_i from course difficulty d_c in the additive model $P_{ic} = a_i - d_c + \varepsilon_{ic}$, and that *eigengrades* — fixed-effect adjustment over the connected enrollment graph — recover the centered student effect $a_i - \bar{a}$ *ex post*, neutralizing the competitive student- and instructor-side channels under a report-only market benchmark. Eigengrades are, however, an *estimator*: they presuppose honest raw grades and carry no incentive-compatibility guarantee. This paper asks whether a VCG/Groves-style *mechanism*, restricted to **report-space instruments with no monetary transfers**, can make informative, difficulty-free transcripts the *equilibrium* outcome of strategic instructors and students. The organizing idea is that fixed-supply report-space goods — enrollment share, aggregate weight, and disclosure priority — can play the role of the VCG numéraire, because eigengrade precision is a global function of the transcript-graph Laplacian spectrum and therefore assigns each agent a well-defined externality to be internalized by a Clarke pivot.

This is a living document maintained by the AI co-mathematician system. Items flagged **[UNPROVEN 1]** are explicitly unverified and listed in the appendix of open obligations.

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Contents

1	Introduction	1
2	Environment	2
2.1	Strategic primitives carried over	2
3	The planner’s objective and the money-VCG benchmark (SG0)	3
3.1	Strategic action and weighted-Laplacian recovery	3
3.2	The externality kernel	3
3.3	The Groves benchmark	3
3.4	Well-posedness of the efficient rule	4
4	Instructor mechanism in report space (SG1)	4
4.1	Global structure of the share game: an exact concave potential (SG1-a)	5
5	Student mechanism in report space (SG2)	8
5.1	The extensive-margin externality is the SG0 kernel	8
5.2	The fixed-budget pseudo-market	9
5.3	Externality pricing and the fixed-budget tax	9
5.4	Efficiency loss vs. the money-VCG benchmark	10

6	Truthful performance revelation (SG3) — PARTIAL / UNDER REVISION	11
6.1	Model (inherits SG0 D1, Gaussian specialization)	11
6.2	Failure modes that are NOT ruled out (retained)	12
7	The no-money boundary: the “α dominates” wall (SG4)	12
7.1	The mechanism class and its single bounded lever	12
7.2	Deriving the intensity cap $\bar{\eta}$ from share feasibility	13
7.3	The implementability threshold and the impossibility	13
7.4	Scope: class-specific vs. fully mechanism-independent	14
8	Two-sided composition: existence and cross-margin interaction (SG5)	15
8.1	The composite game	15
8.2	The two sides are concave games	16
8.3	Existence for the composite game	16
8.4	Cross-margin interaction: how student connectivity moves the fidelity incentive	17
8.5	What does <i>not</i> compose: open obligations	18
8.6	Computational validation on the running example	18
9	Discussion	18
A	Open obligations	19
B	Failed explorations (negative space)	22

1 Introduction

Gans and Kominers (2026) solve a *measurement* problem: a scalar grade confounds student ability a_i with course difficulty d_c , and their *eigengrade*—fixed-effect adjustment over the connected enrollment graph—recovers the centered ability $a_i - \bar{a}$ *ex post*, neutralizing the competitive student- and instructor-side channels under a report-only market benchmark. But an eigengrade is an estimator: it presupposes honest raw grades, and it makes informative, difficulty-free transcripts *neutral* rather than *optimal*—students and instructors are given no incentive to supply the overlap and within-course fidelity the estimator consumes.

We ask the mechanism-design question one layer up: can honest, variation-preserving grading and efficient, identification-preserving course choice be made an *equilibrium*, using only **report-space instruments and no monetary transfers**? The organizing object is the precision bound of §2: eigengrade error is a global function of the transcript-graph Laplacian spectrum, so each agent’s report imposes a well-defined externality $\|\Pi L^+ b_{ic}\|^2$ —an effective-resistance form—on everyone else’s measured ability. This *one kernel* prices both margins of manipulation: the *intensive* margin (how faithfully an instructor grades, §4) and the *extensive* margin (which courses a student takes, §5).

Our results, organized as a suite. §3 states the money-VCG *benchmark* (DSIC and informationally efficient *with* transfers) as the frontier the money-free mechanisms are measured against. §4 gives the instructor mechanism: a fixed-supply enrollment-share pivot that reproduces the money pivot up to a **simplex tax** $1 - 1/m$ —the same zero-sum factor as Gans–Kominers’ competitive-inflation term (their eq. 19)—and §4.1 proves the share game is an *exact concave potential game*, delivering global best response and (under coercivity) a unique equilibrium for arbitrary, heterogeneous instructors. §5 gives the student mechanism: a fixed-budget pseudo-market pricing the connectivity externality, incurring the parallel tax $1 - 1/K$ across K students. §6 (truthful performance revelation) is *partial*: the peer-prediction route is reduced, by adversarial review, to a correct independence result plus an open propriety conjecture—we report the negative outcome honestly. §7 proves the project’s own impossibility: the “ α dominates”

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wall, under which no fixed-supply share mechanism can implement full fidelity once the direct grade-preference is strong—the no-money analogue of Gans–Kominers’ Thm. 4.5. §8 composes the two sides, proving equilibrium existence and signing the cross-margin interaction, and validates the suite numerically on the paper’s running example (reproducing the 0.996-vs-0.812 eigengrade-vs-GPA correlation exactly). Throughout, items flagged **[UNPROVEN 2]** are explicitly unverified and collected in Appendix A; the recurring theme is that a fixed-supply report-space numéraire prices an externality only up to a $1 - 1/(\#\text{agents})$ pivot, and is money-equivalent only in the large-economy limit.

2 Environment

We adopt the paper’s scalar additive benchmark verbatim; nothing here is new and the results of **Gans--Kominers** are taken as given infrastructure (see `Out of scope` in `goals.md`).

- Students $S = \{1, \dots, n\}$, courses $C = \{1, \dots, m\}$. Student i has ability $a_i \in \mathbb{R}$; course c has difficulty $d_c \in \mathbb{R}$.
- Latent performance $P_{ic} = a_i - d_c + \varepsilon_{ic}$, $\mathbb{E}[\varepsilon_{ic}] = 0$.
- An observed enrollment graph $\mathcal{G} = (S, C, E)$, $E \subseteq S \times C$, assumed connected (paper Prop. 6.7).
- Signed incidence matrix $B \in \mathbb{R}^{|E| \times (n+m)}$, Laplacian $L = B^\top B$, pseudoinverse L^+ . The eigengrade is $\hat{a}^{\text{GL}} = \text{student block of } N_s(L^+ B^\top p)$, recovering $a - \bar{a} \mathbf{1}_n$ (paper Thm. 6.9).
- **Precision identity (the externality source)**. With $p = Bx^* + \varepsilon$, $\text{Cov}(\varepsilon) = \sigma^2 I$,

$$\mathbb{E} \|\tilde{a} - (a - \bar{a} \mathbf{1}_n)\|^2 \leq \sigma^2 \text{tr}(L^+) \leq \sigma^2 \frac{n+m-1}{\lambda_2(L)},$$

where $\lambda_2(L) > 0$ is the algebraic connectivity (paper Prop. 6.15). Every agent’s report or course choice perturbs L (and the per-course noise σ_c^2 via within-course variation), hence perturbs the recovered ability of *all* students. This shared-precision term is the externality a report-space VCG mechanism must price.

2.1 Strategic primitives carried over

Student course-choice utility (paper eq. 13): $U_{ic} = v_{ic} + \gamma m_i(c; \mathcal{R}) + \nu_{ic}$, with strategic susceptibility $\Omega(\mathcal{R})$ (eq. 12). Instructor payoff (paper eq. 18): $U_c = \alpha \bar{G}_c + \beta \log s_c(\bar{G}; \mathcal{R}) - \frac{\xi}{2}(\bar{G}_c - \bar{G}_0)^2$, with the share equation s_c on the simplex (eq. 16) and competitive-inflation term $\beta \eta_{\mathcal{R}}(1 - 1/m)/\xi$. The simplex structure of s_c — a *fixed-supply* good — is what we will exploit as the money-free numéraire.

3 The planner’s objective and the money-VCG benchmark (SG0)

This benchmark is the efficiency frontier the no-money mechanisms of §§4–6 are measured against. The non-routine content is the explicit form of the informativeness objective and of each agent’s externality in terms of the *weighted* transcript-graph Laplacian; the incentive logic is then standard Groves.

3.1 Strategic action and weighted-Laplacian recovery

Exact difficulty adjustment already neutralises a uniform course-level shift (Gans–Kominers Cor. 8.4), so the live margin is *within-course fidelity*: instructor c injects reporting noise of variance $\tau_c^2 \geq 0$ (compression, coarse curving, distortion), reporting $g_{ic} = P_{ic} + h_c + \zeta_{ic}$, $\zeta_{ic} \sim (0, \tau_c^2)$, with h_c an irrelevant level shift. The institution runs weighted GLS with edge precisions $w_{ic} = (\sigma^2 + \tau_c^2)^{-1}$; let $W = \text{diag}(w_{ic})$, $L_W = B^\top W B$, and $\hat{x} = L_W^+ B^\top W p$.

Lemma 3.1 (GLS covariance). *With per-edge noise covariance W^{-1} , $\text{Cov}(\hat{x}) = L_W^+$.*

Proof. $\hat{x} = H p$, $H = L_W^+ B^\top W$; then $\text{Cov}(\hat{x}) = H W^{-1} H^\top = L_W^+ B^\top W B L_W^+ = L_W^+ L_W L_W^+ = L_W^+$. $\square$

On the connected graph $\ker(L_W) = \text{span}\{\mathbf{1}_{n+m}\}$, so the zero-sum-normalised student block $\tilde{\alpha} = M_n \hat{\alpha}$ is unbiased for $a - \bar{a} \mathbf{1}_n$ (Gans–Kominers Prop. 6.15, σ^2 folded into W). With $\Pi = \text{diag}(M_n, 0_m)$ (symmetric idempotent), define the **informativeness welfare**

$$W(\tau) = -\mathbb{E}\|\tilde{\alpha} - (a - \bar{a} \mathbf{1}_n)\|^2 = -\text{tr}(\Pi L_W^+ \Pi), \quad w_{ic} = (\sigma^2 + \tau_c^2)^{-1}.$$

3.2 The externality kernel

Proposition 3.2 (Edge-precision gradient). *For every edge e , $\frac{\partial W}{\partial w_e} = \|\Pi L_W^+ b_e\|^2 \geq 0$, and consequently $\frac{\partial W}{\partial \tau_c} = -\frac{2\tau_c}{(\sigma^2 + \tau_c^2)^2} \sum_{i:(i,c) \in E} \|\Pi L_W^+ b_{ic}\|^2 \leq 0$.*

Proof. $\partial L_W / \partial w_e = b_e b_e^\top$. In the Golub–Pereyra differential of L_W^+ , the two pseudoinverse-correction terms carry the factor $(\partial L_W)(I - L_W L_W^+) = b_e (b_e^\top \mathbf{1}) \mathbf{1}^\top / (n + m)$, which vanishes because each incidence row satisfies $b_e^\top \mathbf{1} = 1 - 1 = 0$. Hence $\partial L_W^+ = -L_W^+ b_e b_e^\top L_W^+$ and $\partial W / \partial w_e = \text{tr}(\Pi L_W^+ b_e b_e^\top L_W^+ \Pi) = \|\Pi L_W^+ b_e\|^2$. The τ_c form uses $\partial w_{ic} / \partial \tau_c = -2\tau_c / (\sigma^2 + \tau_c^2)^2$. $\square$

The quantity $\|\Pi L_W^+ b_{ic}\|^2$ is an effective-resistance-type form: it is the **measurement-precision externality** instructor c ’s compression imposes on the recovered ability of every student reachable through course c . It is the object the no-money mechanisms must internalise.

3.3 The Groves benchmark

Instructor c has private type θ_c and intrinsic payoff $u_c(\tau_c; \theta_c)$ (a taste for lenient/compressed grading — the residual α channel of Gans–Kominers §8). Define total welfare $\mathcal{W}(\tau; \theta) = W(\tau) + \sum_c u_c(\tau_c; \theta_c)$. In the direct mechanism, instructors report $\hat{\theta}$; the planner picks $\tau(\hat{\theta}) \in \arg \max_\tau \mathcal{W}(\tau; \hat{\theta})$ and Groves transfers $t_c(\hat{\theta}) = W(\tau(\hat{\theta})) + \sum_{k \neq c} u_k(\tau_k; \hat{\theta}_k) + h_c(\hat{\theta}_{-c})$.

Theorem 3.3 (Money-VCG benchmark). *Under the efficient rule with Groves transfers, truthful reporting $\hat{\theta}_c = \theta_c$ is a dominant strategy for every instructor, and the induced fidelity profile maximises total welfare $\mathcal{W}(\cdot; \theta)$.*

Proof. Instructor c ’s payoff is $u_c(\tau_c(\hat{\theta}); \theta_c) + t_c = \mathcal{W}(\tau(\hat{\theta}); (\theta_c, \hat{\theta}_{-c})) + h_c(\hat{\theta}_{-c})$, and h_c is independent of $\hat{\theta}_c$. Reporting truthfully makes the planner maximise exactly the instructor’s objective, for any $\hat{\theta}_{-c}$; hence truth is dominant and the outcome is efficient at the true profile. $\square$

Corollary 3.4 (Pure informativeness). *If $u_c \equiv 0$ (or transfers are scaled so W dominates), the efficient rule sets $\tau_c = 0$ for all c (Prop. 3.2), implementing full-fidelity, maximally informative grading in dominant strategies.*

The mechanism works *only* through the transfer t_c , which pays instructor c the social value $W + \sum_{k \neq c} u_k$ in a transferable numéraire. The program of §§4–6 is to reproduce the marginal incentive $\partial t_c \propto \partial W / \partial \tau_c$ using fixed-supply report-space goods instead; §7 bounds what is lost.

3.4 Well-posedness of the efficient rule

The efficient rule is well-posed, but the relevant curvature lives in *precision* space, not compression space.

Proposition 3.5 (Concavity in precision; failure in compression). *On the connected graph, $F(w) = \text{tr}(\Pi L_W^+ \Pi)$ is convex in the edge-precision vector $w \in (0, \infty)^{|E|}$, so $W = -F$ is concave in w . Writing $w_{ic} = 1/(\sigma^2 + \tau_c^2)$, the welfare $W_\tau(\tau) = W(w(\tau))$ is concave on the low-compression box $\{0 \leq \tau_c \leq \sigma/\sqrt{3} \forall c\}$ but is not concave on $[0, \infty)^m$ in general.*

Proof. Let U have orthonormal columns spanning $\mathbf{1}^\perp$. As every $b_e \in \mathbf{1}^\perp$, $A(w) = U^\top L_W U = \sum_e w_e (U^\top b_e)(U^\top b_e)^\top$ is affine in w and positive definite (if $x^\top A x = 0$ then Ux is constant on every edge, so by connectivity $Ux \in \text{span}\{\mathbf{1}\} \cap \mathbf{1}^\perp = \{0\}$). Then $L_W^+ = U A(w)^{-1} U^\top$ and, with $C = U^\top \Pi^2 U \succeq 0$, $F(w) = \text{tr}(C A(w)^{-1})$. The inverse is operator-convex (Löwner) and $\text{tr}(C \cdot)$ with $C \succeq 0$ preserves the Löwner order, so F is convex along the affine map $w \mapsto A(w)$. For the τ claims, $g(t) = 1/(\sigma^2 + t^2)$ satisfies $g''(t) = 2(3t^2 - \sigma^2)/(\sigma^2 + t^2)^3$, so g is concave on $[0, \sigma/\sqrt{3}]$; a concave-nondecreasing function (W in w) of coordinatewise-concave maps is concave, giving concavity on the box. Global failure: on $K_{2,2}$ with $\sigma^2 = 1$, $W_\tau(\tau_1, \tau_2) = -(\frac{1}{1+\tau_1^2} + \frac{1}{1+\tau_2^2})^{-1}$, and along $\tau_2 = 0$, $W_\tau(t, 0) = -(1+t^2)/(2+t^2)$ has $\partial_t^2 > 0$ for $t > \sqrt{2/3}$; e.g. $W_\tau(\frac{3}{2}, 0) = -\frac{13}{17} < -\frac{31}{44} = \frac{1}{2}[W_\tau(0, 0) + W_\tau(3, 0)]$. $\square$

Remark 3.6 (What DSIC actually needs). Theorem 3.3 requires *no* concavity: Groves holds for any efficient selection rule with a fixed tie-break. Concavity (Prop. 3.5, in w) matters only for uniqueness and comparative statics. Existence of $\arg \max_\tau [W_\tau + \sum_c u_c]$ holds by continuity plus coercivity; via the bound $F(w) \geq x^\top \Pi^2 x / x^\top L_W x$ for $x \in \mathbf{1}^\perp$, a vanishing direction forces $F \rightarrow \infty$ iff it admits $x \in \ker(L^*) \cap \mathbf{1}^\perp$ with $\Pi x \neq 0$ — i.e. the lost edges separate *students* into ≥ 2 components; a peeling-off leaf course leaves F bounded. Absent such a direction, coercivity must come from u_c or from compactifying the action set $\tau_c \in [0, \bar{\tau}]$.

Open obligations (this section). **[UNPROVEN 3]** Individual rationality of the Clarke-pivot ($h_c = -\max_\tau [W + \sum_{k \neq c} u_k]$) version: instructor participation when the pivot charge $-t_c \leq 0$. [ASSUMPTION A-COERC: compactify $\tau_c \in [0, \bar{\tau}]$ or assume coercive u_c where the precision objective alone is non-coercive.] [DECISIONS: additive-noise model of compression (D1); utilitarian vs. pure-informativeness target (D2) — see W002 report.]

4 Instructor mechanism in report space (SG1)

The numéraire is **enrollment share** — fixed-supply ($\sum_c s_c = 1$), already valued by instructors (the β term of Gans–Kominers eq. 18), and, because exact adjustment severs share from mean grades (their Cor. 8.4), free to be re-attached to *fidelity*. The institution observes within-course dispersion and sets shares by a softmax compression penalty $s_c(\tau) = \exp\{-\eta\psi_c(\tau_c)\} / \sum_k \exp\{-\eta\psi_k(\tau_k)\}$, $\psi'_c > 0$. Instructor c 's payoff is $V_c(\tau) = u_c(\tau_c) + \beta \log s_c(\tau)$ with $u'_c > 0$ (taste for compression — the reduced form of the residual α channel).

Proposition 4.1 (Share-pivot reproduces the money-pivot, taxed by $1 - 1/m$). *For the softmax rule, instructor c 's first-order condition is $u'_c(\tau_c) = \beta \eta \psi'_c(\tau_c)(1 - s_c(\tau))$. In a course-symmetric environment, the utilitarian-efficient symmetric profile $\tau^* \mathbf{1}$ (where $u'(\tau^*) = -\partial_{\tau_c} W(\tau^* \mathbf{1})$) is a symmetric Nash equilibrium of the share game under intensity*

$$\eta^* = \frac{u'(\tau^*)}{\beta(1 - 1/m)\psi'(\tau^*)}, \quad \text{equivalently} \quad \underbrace{\beta \eta^* \psi'(\tau^*)(1 - \frac{1}{m})}_{\text{share-pivot marginal}} = \underbrace{-\partial_{\tau_c} W(\tau^* \mathbf{1})}_{\text{money-pivot marginal (SG0)}}.$$

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Status. The FOC matching is exact algebra. The *global* best-response and uniqueness steps—left open in the first iteration—are discharged in §4.1 (Theorem 4.6): the share game is an exact *concave* potential game, so (FOC_c) characterizes the global best response and (under the coercivity hypothesis flagged there) the equilibrium is the unique maximizer of one scalar potential, for arbitrary heterogeneous instructors. The potential turns out to contain *no* welfare term W (Rem. 8.4), correcting the form first conjectured here. (Numerically confirmed on $K_{5,4}$: efficient $\tau^* = 0.556$, $\eta^* = 0.667$, instructor best response = 0.555, both pivot marginals = 0.278.)

Remark 4.2 (The simplex tax). The factor $1 - 1/m$ is the zero-sum cost of a fixed-supply numéraire: one instructor's share gain is the others' loss, so the private marginal value of fidelity is the $(1 - 1/m)$ fraction of its social value, and the institution must inflate intensity by $m/(m - 1)$ to match money. This is exactly the $(1 - 1/m)$ of Gans–Kominers' competitive-inflation term $\beta\eta\mathcal{R}(1 - 1/m)/\xi$ (their eq. 19) — there the strength of the inflation race, here the strength of the fidelity-restoring pivot. The numéraire is money-equivalent only as $m \rightarrow \infty$.

Remark 4.3 (Interior optimum sharpens the target). Since $\partial_{\tau_c} W \propto \tau_c$ vanishes at $\tau = 0$, the utilitarian optimum is interior ($\tau^* > 0$): a little compression is first-order valuable to instructors and only second-order costly to informativeness. The *pure-informativeness* target $\tau = 0$ is a corner, a Nash equilibrium iff $u'(0) \leq \beta\eta\psi'(0)(1 - 1/m)$; since intensity is bounded ($\eta \leq \bar{\eta}$, else winner-take-all shares break low-share IR and the student assignment), full fidelity is implementable iff the direct grade-preference is weak, $u'(0) \leq \beta\bar{\eta}\psi'(0)(1 - 1/m)$. **[UNPROVEN 4]** The " α dominates" impossibility: formalize $\bar{\eta}$ from an IR / assignment-feasibility constraint and prove non-implementability of $\tau = 0$ above the threshold (bridge to §7).

4.1 Global structure of the share game: an exact concave potential (SG1-a)

Proposition 4.1 matched first-order conditions; it left open **[UNPROVEN 5]** **[RESOLVED BELOW, MODULO THE COERCIVITY HYPOTHESIS (H-COERC)]** the global best-response step of the share game in general (asymmetric instructors included). We close the *global best-response* and *uniqueness* parts unconditionally by exhibiting an *exact concave potential*; *existence* of an equilibrium is recovered under the same coercivity/compactness condition already required in §3 (Assumption A-coerc), and we flag exactly where it is needed.

Throughout, instructor $c \in \{1, \dots, m\}$ chooses $\tau_c \geq 0$ with payoff

$$V_c(\tau) = u_c(\tau_c) + \beta \log s_c(\tau), \quad s_c(\tau) = \frac{e^{-\eta\psi_c(\tau_c)}}{\sum_k e^{-\eta\psi_k(\tau_k)}},$$

$\beta, \eta > 0$, with each $u_c, \psi_c \in C^2$, $u'_c > 0$, $\psi'_c > 0$. Write the *log-partition*

$$\Lambda(\tau) = \log \sum_{k=1}^m e^{-\eta\psi_k(\tau_k)}, \quad \text{so} \quad \log s_c(\tau) = -\eta\psi_c(\tau_c) - \Lambda(\tau).$$

Proposition 4.4 (The share game is an exact potential game). *The function*

$$\Phi(\tau) = \sum_{c=1}^m [u_c(\tau_c) - \beta\eta\psi_c(\tau_c)] - \beta\Lambda(\tau)$$

is an exact potential for the share game: for every c and every $\tau \in [0, \infty)^m$,

$$\frac{\partial \Phi}{\partial \tau_c}(\tau) = \frac{\partial V_c}{\partial \tau_c}(\tau) = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c)(1 - s_c(\tau)).$$

Proof. The softmax derivative is $\partial_{\tau_c} \Lambda = (\sum_k e^{-\eta\psi_k})^{-1} \cdot (-\eta\psi'_c(\tau_c))e^{-\eta\psi_c} = -\eta\psi'_c(\tau_c) s_c(\tau)$. Hence

$$\frac{\partial V_c}{\partial \tau_c} = u'_c(\tau_c) + \beta \partial_{\tau_c} (-\eta\psi_c(\tau_c) - \Lambda) = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c) + \beta\eta\psi'_c(\tau_c)s_c = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c)(1 - s_c),$$

which is exactly (FOC_c) of Proposition 4.1. On the other hand, since only the c -th summand of $\sum_k [u_k - \beta\eta\psi_k]$ depends on τ_c ,

$$\frac{\partial\Phi}{\partial\tau_c} = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c) - \beta\partial_{\tau_c}\Lambda = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c) + \beta\eta\psi'_c(\tau_c)s_c = \frac{\partial V_c}{\partial\tau_c}. \quad \square$$

Because Φ is an explicit C^2 scalar function whose gradient reproduces every private marginal incentive, the share game is an *exact* potential game (Monderer–Shapley): the entire stationary structure of the game is the stationary structure of the single scalar Φ . The fact doing the work is that instructors are coupled *only* through the common scalar Λ , and $\partial_{\tau_c}\Lambda$ carries the same factor $\eta\psi'_c(\tau_c)s_c$ that the softmax injects into each private payoff; this is the report-space analogue of the logit-congestion potential and mirrors Gans–Kominers’ potential treatment of the mean-grade game (their Prop. 8.3).

Lemma 4.5 (Hessian of the potential). *On $[0, \infty)^m$,*

$$\nabla^2\Phi(\tau) = \underbrace{\text{diag}(u''_c(\tau_c))}_{\text{(I)}} - \underbrace{\beta\eta \text{diag}(\psi''_c(\tau_c)(1 - s_c))}_{\text{(II)}} - \underbrace{\beta D(\text{diag}(s) - ss^\top)D}_{\text{(III)}},$$

where $s = s(\tau)$ and $D = \text{diag}(g'_c)$ with $g'_c = -\eta\psi'_c(\tau_c)$.

Proof. Differentiate $\partial_{\tau_c}\Phi = u'_c(\tau_c) - \beta\eta\psi'_c(\tau_c)(1 - s_c)$. The diagonal contribution of the u - and ψ -level terms gives u''_c and, from $-\beta\eta\psi''_c(\tau_c)(1 - s_c)$, term (II). It remains to differentiate $+\beta\eta\psi'_c(\tau_c)s_c$ in τ_b . With $g_c = -\eta\psi'_c(\tau_c)$, the softmax Jacobian is the standard $\partial s_c/\partial g_b = s_c(\delta_{cb} - s_b)$, and $\partial g_b/\partial \tau_b = -\eta\psi'_b(\tau_b) = g'_b$, so

$$\frac{\partial}{\partial \tau_b}(\beta\eta\psi'_c(\tau_c)s_c) = \beta\eta\psi'_c(\tau_c)s_c(\delta_{cb} - s_b)g'_b = -\beta g'_c s_c(\delta_{cb} - s_b)g'_b,$$

using $\eta\psi'_c(\tau_c) = -g'_c$. The matrix $[s_c(\delta_{cb} - s_b)]_{c,b} = \text{diag}(s) - ss^\top$, so this block equals $-\beta D(\text{diag}(s) - ss^\top)D$, i.e. $-(\text{III})$. Adding the three pieces gives the claim. (The Hessian is symmetric, as it must be for an exact potential: $\text{diag}(s) - ss^\top$ is symmetric and D is diagonal; equivalently the gradient field $(\partial_{\tau_c}V_c)_c$ has a symmetric Jacobian, confirmed numerically to 5×10^{-10} .) $\square$

Theorem 4.6 (Concavity, global best response, and uniqueness). *Suppose each u_c is concave and each ψ_c is convex on $[0, \infty)$. Then:*

- (i) Φ is concave on $[0, \infty)^m$; if in addition $u''_c < 0$ for every c , Φ is strictly concave.
- (ii) Each instructor’s payoff $V_c(\cdot, \tau_{-c})$ is concave in its own τ_c , so the FOC (FOC_c), together with the boundary comparison at $\tau_c = 0$ and at the upper end of the admissible interval, characterises the global best response; the local matching of Proposition 4.1 is therefore the true global best response, not merely a critical point.
- (iii) Equilibria of the share game are exactly the (coordinatewise-KKT) maximizers of Φ over the admissible action set. Under strict concavity such a maximizer is unique when it exists.

Proof. (i) *Concavity.* We show each of (I), (II), (III) in Lemma 4.5 is negative semidefinite, so $\nabla^2\Phi = (\text{I}) - (\text{II}) - (\text{III})$ is NSD. (I) = $\text{diag}(u''_c) \preceq 0$ since each u_c is concave. (II) = $\beta\eta \text{diag}(\psi''_c(1 - s_c)) \succeq 0$ because $\beta, \eta > 0$, $\psi''_c \geq 0$ (convex ψ_c) and $1 - s_c \in (0, 1)$; hence $-(\text{II}) \preceq 0$. For (III), the matrix $\Sigma := \text{diag}(s) - ss^\top$ is the covariance of the categorical indicator e_X ($\Pr[X = c] = s_c$): $\mathbb{E}[e_X e_X^\top] = \text{diag}(s)$, $\mathbb{E}[e_X]\mathbb{E}[e_X]^\top = ss^\top$, so $\Sigma = \text{Cov}(e_X) \succeq 0$, whence $D\Sigma D \succeq 0$ for diagonal D and $-(\text{III}) \preceq 0$. Summing, $\nabla^2\Phi \preceq 0$ on all of $[0, \infty)^m$. If every $u''_c < 0$ then (I) $\prec 0$ and $\nabla^2\Phi \preceq (\text{I}) \prec 0$, so Φ is strictly concave.

(ii) *Global own-concavity.* Fixing τ_{-c} , $\partial_{\tau_c}^2 V_c = [\nabla^2 \Phi]_{cc} = u_c'' - \beta \eta \psi_c''(1 - s_c) - \beta (g_c')^2 s_c(1 - s_c) \leq 0$, each summand ≤ 0 under the hypotheses. Thus $V_c(\cdot, \tau_{-c})$ is concave on the admissible interval, so the global maximizer is identified by the interior root of (FOC_c) when one exists, and otherwise by the relevant boundary; in particular the best response computed in Proposition 4.1 is the true global best response.

(iii) *Equilibria as KKT points of a concave program.* By Proposition 4.4 and own-concavity (ii), τ is a Nash equilibrium iff, for every c , $\partial_{\tau_c} \Phi(\tau) \leq 0$ with equality when τ_c is interior (and the matching boundary complementarity at the upper end, when the action set is bounded above). These coordinatewise KKT conditions are exactly the KKT conditions of $\max \Phi$ over the admissible set; since Φ is concave (i) they are necessary and sufficient for a global maximizer. Hence the equilibrium set equals the maximizer set, and under strict concavity the maximizer—when it exists—is unique. $\square$

Remark 4.7 (Existence requires coercivity: a flagged gap). Theorem 4.6 gives concavity, global best response and (strict case) at-most-one equilibrium *unconditionally*, but it does *not* by itself give existence. The hypotheses “ u_c concave, ψ_c convex” do *not* make Φ coercive on $[0, \infty)^m$: **[UNPROVEN 6]** [REVIEWER COUNTEREXAMPLE] take $m = 2$, $\beta = \eta = 1$, $u_c(t) = 2t$ (concave, $u_c' = 2 > 0$) and $\psi_c(t) = 0.1t$ (convex, $\psi_c' = 0.1 > 0$). Then along the diagonal $\Phi(t, t) = 1.9t - \beta\Lambda$ diverges to $+\infty$, each $V_c(\cdot, \tau_{-c})$ is strictly increasing in its own argument, and the share game has *no* Nash equilibrium on $[0, \infty)^m$. Thus existence is genuinely conditional and is hereby flagged, not proved, in the unbounded setting. Two clean ways to recover existence:

- **(Box / A-coerc).** Restrict to the compact action box $\tau_c \in [0, \bar{\tau}]$ of Assumption A-coerc (§3). Then Φ is continuous on a compact set and attains its maximum, which by Theorem 4.6(iii) is the unique (strict case) equilibrium *of the box game*. Caveat: if the maximizer lies at the upper boundary $\tau_c = \bar{\tau}$ with $\partial_{\tau_c} \Phi > 0$, it is an equilibrium of the box game only, not of the unconstrained game on $[0, \infty)^m$.
- **(H-coerc).** Assume $u_c(t) - \beta \eta \psi_c(t) \rightarrow -\infty$ as $t \rightarrow \infty$ for every c (e.g. ψ_c superlinear, or u_c bounded above with ψ_c unbounded). Then $\Phi \rightarrow -\infty$ as $\|\tau\| \rightarrow \infty$, the maximizer exists and is interior-from-above in every coordinate, and the equilibrium set equals $\arg \max_{\tau \geq 0} \Phi$ on the full orthant. The quadratic- ψ family used in the numerics ($\psi_c(t) = \frac{1}{2} P_c t^2$) satisfies H-coerc, which is why a unique interior equilibrium is observed there.

Remark 4.8 (What this buys; the asymmetric case). Under either H-coerc or A-coerc, Theorem 4.6 closes the global-best-response and uniqueness gap of Proposition 4.1 *for arbitrary, heterogeneous instructors* (u_c, ψ_c): the equilibrium is the unique maximizer of one concave program Φ , which the institution reaches by announcing η . *Interiority* is a further, conditional property: the equilibrium can be the corner $\tau = 0$ (a Nash equilibrium iff $u_c'(0) \leq \beta \eta \psi_c'(0)(1 - s_c)$ for all c , cf. Remark 4.3) rather than interior; we do *not* claim an interior equilibrium unconditionally. What *course-symmetry* buys is separate: matching the equilibrium to the *utilitarian-efficient* profile τ^* . With a single scalar η the designer has one degree of freedom but m social first-order conditions $u_c'(\tau_c^*) = -\partial_{\tau_c} W(\tau^*)$; in the symmetric environment these collapse to one and η^* of Proposition 4.1 solves it exactly. In an asymmetric environment the mechanism still pins a unique equilibrium (given coercivity), but generically *not* the efficient one; to recover efficiency the designer uses the m course-specific penalties ψ_c (equivalently per-course η_c folded into ψ_c), choosing $\psi_c'(\tau_c^*)$ so each (FOC_c) coincides with the social FOC at τ^* . Thus the open “one η cannot match m FOCs” worry is an *efficiency-matching* issue, not a best-response/uniqueness obstruction (those hold under coercivity), and is closed by enriching the penalty profile. **[UNPROVEN 7]** Existence of a pure Nash equilibrium of the share game absent any coercivity/box hypothesis (false in general, by Remark 4.7); the downstream two-sided composition (SG5) should invoke A-coerc or H-coerc rather than the unconditional claim.

5 Student mechanism in report space (SG2)

SG1 priced the *intensive* margin (how faithfully an instructor reports). SG2 prices the *extensive* margin: *which edges of the enrollment graph exist*. A student who chooses a course that thins overlap shrinks the algebraic connectivity $\lambda_2(L)$ and, through the precision bound $\mathbb{E}\|\tilde{a} - (a - \bar{a}\mathbf{1}_n)\|^2 \leq \sigma^2 \text{tr}(L^+) \leq \sigma^2(n + m - 1)/\lambda_2(L)$ of §2, degrades the recovered ability of *every* student. Under exact difficulty adjustment Gans–Kominers’ neutrality result gives students no private reason to internalise this: their course-choice utility is pedagogical match plus idiosyncratic taste, invariant to the global spectral consequence. With money this is a Groves *subsidy* (efficient, by the same logic as Theorem 3.3). The content of SG2 is to reproduce that subsidy with a fixed-supply, money-free numéraire, and to be honest about the residual gap.

5.1 The extensive-margin externality is the SG0 kernel

Fix a connected *spine* of mandatory enrollments and let each student i choose an elective $c(i)$ from a menu $\mathcal{M}_i \subseteq C$; write the chosen edge set $E(\mathbf{c}) = E_0 \cup \{(i, c(i))\}_i$ and Laplacian $L(\mathbf{c})$ (unit edge weights here; the SG1 precisions w compose multiplicatively and are suppressed). Student i ’s pedagogical value of seat c is $v_{ic} \geq 0$; the planner’s pedagogical objective is $\text{Ped}(\mathbf{c}) = \sum_i v_{ic(i)}$.

Proposition 5.1 (Extensive margin is priced by the same kernel). *Treat an edge (i, c) as carried at weight w_{ic} and let $W(\mathbf{c}) = -\text{tr}(\Pi L^+ \Pi)$ be the informativeness welfare of §3. Then the marginal informativeness value of strengthening that edge is*

$$\frac{\partial W}{\partial w_{ic}} = \|\Pi L^+ b_{ic}\|^2 \geq 0,$$

the same kernel that prices the instructor intensive margin (Prop. 3.2). Hence adding an edge weakly raises W , and the seat a student fills carries a nonnegative connectivity value $\rho_{ic} := \|\Pi L^+ b_{ic}\|^2$.

Proof. Identical to Prop. 3.2: $b_{ic}^\top \mathbf{1} = 0$ kills the pseudoinverse-correction terms in the Golub–Pereyra differential, leaving $\partial L^+ = -L^+ b_{ic} b_{ic}^\top L^+$ and $\partial W / \partial w_{ic} = \|\Pi L^+ b_{ic}\|^2$. Nonnegativity is immediate. $\square$

(Verified: central finite-difference $\partial W / \partial w_e = 0.37500000$ matches the kernel 0.37500000; see C1 of the verification output.)

Remark 5.2 (Connectivity, not just resistance). The relevant scalar is global: $\text{tr}(L^+) = \sum_{k \geq 2} 1/\lambda_k(L)$ is dominated by $1/\lambda_2$. A choice that disconnects the *students* into ≥ 2 components sends $\text{tr}(\Pi L^+ \Pi) \rightarrow \infty$ (Remark 3.6); the connectivity value ρ_{ic} is the smooth, edge-local proxy for this global identification risk.

5.2 The fixed-budget pseudo-market

We use an artificial-currency Fisher market in the Hylland–Zeckhauser [2] / Budish [3] tradition. Each student receives an equal, fixed budget of $B > 0$ *priority points*, a non-transferable scrip that buys nothing outside the assignment (the report-space stand-in for money). A (possibly fractional) allocation is a doubly capacitated point of the assignment polytope

$$\Delta = \left\{ x \in \mathbb{R}_{\geq 0}^{S \times C} : \sum_c x_{ic} = 1 \ \forall i, \ \sum_i x_{ic} \leq \text{cap}_c \ \forall c, \ x_{ic} = 0 \text{ if } c \notin \mathcal{M}_i \right\},$$

with x_{ic} the probability student i holds seat c . The planner posts **seat prices proportional to the marginal connectivity externality**,

$$q_c = \bar{q} - \kappa \bar{\rho}_c, \quad \bar{\rho}_c = \frac{1}{|\{i: c \in \mathcal{M}_i\}|} \sum_{i: c \in \mathcal{M}_i} \rho_{ic}, \quad \kappa \geq 0,$$

i.e. overlap-creating seats are *rebated* (made cheaper) in proportion to the informativeness they restore; $\bar{q}$ and κ are the two planner dials. Each student spends their fixed budget to maximise pedagogical match net of price,

$$\max_{x_i \in \Delta_i} \sum_c x_{ic} (v_{ic} + \kappa \rho_{ic}) \quad \text{s.t.} \quad \sum_c q_c x_{ic} \leq B,$$

and the planner adjusts $(\bar{q}, \kappa)$ to clear seats and meet a **connectivity floor** $\lambda_2(L(\mathbf{c})) \geq \lambda_{\min}$.

Proposition 5.3 (Existence of a pseudo-market equilibrium). *With equal fixed budgets B , strictly positive seat prices $q_c > 0$, and the compact convex feasible set Δ , a competitive (Fisher) equilibrium (x^*, q^*) of the fractional assignment economy exists: x_i^* is budget-feasible and optimal for each i at prices q^* , and markets clear ($\sum_i x_{ic}^* \leq \text{cap}_c$, with equality where $q_c^* > 0$).*

Proof. Each student's demand correspondence $D_i(q) = \arg \max \{ \sum_c x_{ic} (v_{ic} + \kappa \rho_{ic}) : x_i \in \Delta_i, q^\top x_i \leq B \}$ is, on the compact convex Δ_i with the continuous linear objective and the continuous budget constraint, nonempty, convex-valued and upper hemicontinuous (Berge). Seat supply is the fixed simplex of capacities. This is a linear Fisher market with a finite divisible good supply and equal incomes; existence of a competitive equilibrium follows from the Eisenberg–Gale convex program $\max_x \sum_i B \log(\sum_c x_{ic} (v_{ic} + \kappa \rho_{ic}))$ over Δ , whose KKT prices q^* support x^* ([4]; the program is concave, Δ is a nonempty compact polytope, so a maximiser and supporting multipliers exist). **[UNPROVEN 8]** The connectivity floor $\lambda_2(L(\mathbf{c})) \geq \lambda_{\min}$ is non-convex in $\mathbf{c}$ (it is a concave spectral function of L , hence the feasible set $\{\lambda_2 \geq \lambda_{\min}\}$ is convex in the edge-weight relaxation but the integral assignment it must round to is not); existence of an equilibrium that simultaneously clears and meets the floor in integral allocations, and a rounding scheme preserving the floor, is not established. We prove existence for the fractional relaxation without the floor, and treat the floor as a post-hoc feasibility filter. $\square$

5.3 Externality pricing and the fixed-budget tax

At equilibrium each student equalises the marginal pedagogical-plus-connectivity value per priority point across the seats they buy. Setting the planner's rebate intensity κ equal to the Groves rate would make the student's private objective $v_{ic} + \kappa \rho_{ic}$ coincide with the social marginal $v_{ic} + \partial W / \partial w_{ic}$ of adding the edge — exactly the Groves connectivity subsidy of the money benchmark. The obstruction is that priority points are *fixed supply* and shared across K competing students, so a unit of rebate posted on one seat is mechanically withdrawn from the pool available to the others.

Proposition 5.4 (Fixed-budget fractional pivot — the SG2 simplex tax). *In the equal-fixed-budget economy with K students drawing on a common, normalised priority-point supply, the connectivity rebate a planner can fund through prices recovers only the fraction*

$$\rho_{\text{tax}} = \frac{K-1}{K} = 1 - \frac{1}{K}$$

of the first-best Groves subsidy: posting rebate $\kappa \bar{\rho}_c$ on seat c raises every other seat's relative price by $\kappa \bar{\rho}_c / (K-1)$ through the fixed budget constraint, so the net cross-sectional incentive is $(1 - 1/K)$ of the gross. To match the money pivot the planner must inflate the rebate intensity by $1/\rho_{\text{tax}} = K/(K-1)$.

Proof sketch. Normalise total priority points to 1, so each budget is $B = 1/K$. A rebate that lowers seat c 's price by δ frees δx_{ic} of budget for student i but, market-clearing the fixed supply, must be matched by an offsetting $+\delta/(K-1)$ on the seats absorbed by the other $K-1$ students (the fixed-supply adding-up identity $\sum_i \sum_c q_c x_{ic} = KB = 1$). The net own-price reduction seen by the marginal student is therefore $\delta(1 - 1/K)$. The bookkeeping is identical to the SG1 share

simplex with $\sum_c s_c = 1$ replaced by $\sum_i B_i = 1$; there the factor was $1 - 1/m$ across m instructors, here it is $1 - 1/K$ across K students. **[UNPROVEN 9]** This is established as a first-order, symmetric-environment adding-up identity (matching the SG1 status). The exact equilibrium price response in an asymmetric pseudo-market, and the claim that $1/\rho_{\text{tax}}$ is the precise (not merely first-order) inflation needed, are not proved. $\square$

Remark 5.5 (One theme, two margins). SG1’s enrollment-share numéraire is fixed-supply across m instructors and incurs the tax $1 - 1/m$ (Remark 4.2); SG2’s priority-point numéraire is fixed-supply across K students and incurs the tax $1 - 1/K$. Both are the same phenomenon: a fixed-supply report-space currency prices an externality only up to a fractional pivot, and is money-equivalent only in the large-economy limit ($m \rightarrow \infty$, $K \rightarrow \infty$). This is the structural signature of money-free implementation in this model, and it is the same $1 - 1/(\text{\#agents})$ that drives the Gans–Kominers competitive-inflation term (their eq. 19).

5.4 Efficiency loss vs. the money-VCG benchmark

Let $\mathbf{c}^{\text{FB}} \in \arg \max_{\mathbf{c}} [\text{Ped}(\mathbf{c}) + W(\mathbf{c})]$ be the first-best (money-VCG, SG0) allocation and $\mathbf{c}^{\text{PM}}$ the pseudo-market equilibrium at rebate intensity κ . Write $\mathcal{W}(\mathbf{c}) = \text{Ped}(\mathbf{c}) + W(\mathbf{c})$.

Proposition 5.6 (Efficiency-loss bound). *The pseudo-market welfare gap decomposes as*

$$\mathcal{W}(\mathbf{c}^{\text{FB}}) - \mathcal{W}(\mathbf{c}^{\text{PM}}) \leq \underbrace{(1 - \rho_{\text{tax}}) \Delta_W}_{\text{fixed-budget tax}} + \underbrace{\epsilon_{\text{round}}}_{\text{frac.} \rightarrow \text{integral}} + \underbrace{\epsilon_{\text{myopia}}}_{\text{price-taking}},$$

where $\Delta_W = W(\mathbf{c}^{\text{FB}}) - W(\mathbf{c}^{\text{ped}}) \geq 0$ is the connectivity value at stake (pedagogy-only vs. first-best), $1 - \rho_{\text{tax}} = 1/K$, ϵ_{round} is the integrality gap of the fractional equilibrium, and ϵ_{myopia} is the gap between the pseudo-market fixed point and the global maximiser of W on the feasible set.

Proof sketch. At the Groves rate the private and social marginals coincide and the gap is the rounding loss alone; Prop. 5.4 scales the recoverable incentive by ρ_{tax} , leaving an un-internalised $(1 - \rho_{\text{tax}})$ fraction of Δ_W ; price-taking students reach a competitive fixed point, not a joint optimiser, contributing ϵ_{myopia} . The three terms are upper bounds on disjoint sources of loss; their sum bounds the total by the triangle inequality over the path $\mathbf{c}^{\text{ped}} \rightarrow \mathbf{c}^{\text{PM}} \rightarrow \mathbf{c}^{\text{FB}}$. **[UNPROVEN 10]** Each of ϵ_{round} and ϵ_{myopia} is named but not bounded in closed form. Concretely, W (equivalently $\text{tr}(\Pi L^+ \Pi)$) and $\lambda_2(L)$ are *not* co-monotone: the verification instance exhibits a pseudo-market fixed point that attains the *maximal* λ_2 (1.382 vs. pedagogy-only 0.272) yet leaves a residual tr-gap (1.362 vs. the global tr-minimiser 1.333). Hence even the no-myopia, no-tax fractional optimum of a λ_2 - or ρ -priced market need not coincide with the $\text{tr}(\Pi L^+ \Pi)$ first-best. Fully characterising first-best implementation without money is therefore *not* achieved here, consistent with the SG4 boundary. $\square$

Remark 5.7 (What the mechanism does deliver). Numerically (C2–C3 of the verification output, a 4-student/3-course instance), the pedagogy-only (neutral) assignment yields $\lambda_2 = 0.272$, $\text{tr}(\Pi L^+ \Pi) = 3.375$; sweeping the rebate intensity κ from 0 the pseudo-market fixed point moves monotonically to $\lambda_2 = 1.382$, $\text{tr}(\Pi L^+ \Pi) = 1.362$ — a $5\times$ improvement in algebraic connectivity and a $2.5\times$ reduction in the eigengrade error bound — at a pedagogical cost of Ped falling from 3.700 to 3.160 (the efficiency tension the planner trades off through κ). The mechanism uses *no* monetary transfer: every instrument (priority points, seat rebates, the connectivity floor) lives in report/assignment space.

Open obligations (this section). **[UNPROVEN 11]** Existence of an *integral* pseudo-market equilibrium meeting the connectivity floor, and a floor-preserving rounding of the fractional

Eisenberg–Gale equilibrium. [UNPROVEN 12] Exact (not first-order) form of the fixed-budget pivot $\rho_{\text{tax}} = 1 - 1/K$ in asymmetric economies, paralleling the open SG1 asymmetric-share step. [UNPROVEN 13] Closed-form bounds on the rounding gap ϵ_{round} and the price-taking myopia gap ϵ_{myopia} ; in particular whether any report-space (no-money) student mechanism can close the $\text{tr}(\Pi L^+ \Pi)$ -vs- λ_2 non-co-monotonicity gap, which is the natural SG4 impossibility hook.

6 Truthful performance revelation (SG3) — PARTIAL / UN- DER REVISION

This section attempts to turn the eigengrade *residual* into the cross-check of a money-free peer-prediction mechanism. The elicitation layer is prior-art peer prediction (Miller–Resnick–Zeckhauser 2005; Prelec 2004; Witkowski–Parkes 2012; Dasgupta–Ghosh 2013; Shnayder–Agarwal–Frongillo–Parkes 2016; Kong–Schoenebeck 2019; proper scores Brier 1950, Savage 1971, Gneiting–Raftery 2007). The intended novelty is that the correlation device is the connected two-way fixed-effect transcript graph and the prior-art leave-one-out eigengrade $\mu^{-e} = \hat{x}_i^{-e} - \hat{x}_{n+c}^{-e}$.

6.1 Model (inherits SG0 D1, Gaussian specialization)

On the connected graph $\mathcal{G}$ each edge $e = (i, c)$ is graded by ≥ 2 independent graders observing

$$s_e^{(k)} = P_{ic} + \omega_e^{(k)}, \quad P_{ic} = a_i - d_c + \varepsilon_{ic} \sim \mathcal{N}(0, \sigma^2),$$

where the latent shock ε_{ic} is shared across co-graders of the same edge (it is a property of the realized performance, not of the observer). *The W005 simulation violated this and is being corrected.*

Proposition 6.1 (Independence of the leave-one-out residual). *μ^{-e} is a function of r_{-e} only, hence under honest others is statistically independent of grader k 's private noise $\omega_e^{(k)}$, and it estimates the mean $a_i - d_c$, not $P_{ic} = a_i - d_c + \varepsilon_{ic}$. Consequently a proper score against μ^{-e} alone elicits the prior mean $\mathbb{E}[P_{ic}]$, not the private signal.*

Proof. L_{-e}, B_{-e}, r_{-e} omit edge e , so μ^{-e} carries no $\omega_e^{(k)}$ or ε_{ic} term; under the i.i.d. model $\text{Cov}(\mu^{-e}, s_e^{(k)}) = 0$. (Numerically, regression slope of μ^{-e} on $s_e^{(k)}$ is $\approx 8 \times 10^{-4}$.) $\square$

[UNPROVEN 14] Strict propriety / honesty as a best response. The submitted claim that the unique expected-score maximizer is the precision-weighted blend $r^* = (\sigma^{-2} s_e^{(k)} + (v_e^{\text{pr}})^{-1} \mu^{-e}) / (\sigma^{-2} + (v_e^{\text{pr}})^{-1})$ is **not established and is false in the verified construction**. Reviewer numerics (same 5×4 instance, $\sigma = 0.7$): the conditional mean $\mathbb{E}[s_e^{(-k)} | s_e^{(k)}, \mu^{-e}]$ loads essentially zero on μ^{-e} (regression coefficients $[0.70, 1.1 \times 10^{-3}, -3 \times 10^{-5}]$ when ε_{ic} is dropped, as in the submitted code; $[0.35, 0.50, -2 \times 10^{-4}]$ when ε_{ic} is correctly shared). In the submitted simulation a *constant* report equal to the prior mean scores -1.130 , strictly beating the claimed honest blend (-1.348); honest reporting is therefore strictly dominated. A correct propriety theorem must (i) use shared ε_{ic} co-grader signals and (ii) derive the Bayes weights from the correct hierarchy (μ^{-e} a prior on $a_i - d_c$, own signal an observation of $P_{ic} = a_i - d_c + \varepsilon_{ic}$), after which the graph prior's weight is near zero absent further structure.

[UNPROVEN 15] Likelihood / Groves / welfare alignment. The claimed identity “honest $\mathbb{E}[\text{Sc}] = -\frac{1}{2} \log(2\pi v_e) - \frac{1}{2} =$ conditional two-way-FE log-likelihood, hence truthful = welfare-maximizing” is vacuous as argued: v_e was set equal to the honest report's own predictive MSE, and $\mathbb{E}[\text{Sc}] = -\frac{1}{2} \log(2\pi v) - \frac{1}{2}$ holds for *any* report whose MSE equals the plugged-in v . The genuinely score-maximizing report (constant prior mean, $\text{MSE} = \sigma^2$) attains a strictly higher

score, so the link “argmax reward = argmax joint likelihood = $-\text{tr}(\Pi L^+ \Pi)$ ” does not hold in the present construction and must be re-derived. unproven #15

6.2 Failure modes that are NOT ruled out (retained)

[UNPROVEN 16] Uninformative and colluding equilibria are not excluded: with the prior re-computed from manipulable reports, an all-compress profile strictly out-scores honesty (-13.83 vs -21.16). [UNPROVEN 17] Common-prior / Bayes–Nash (not dominant-strategy) dependence; a detail-free correlated-agreement variant on the graph is conjectured but unproved. [UNPROVEN 18] At least two independent signals per edge required (Prop. 6.1); minimal multi-grader coverage open. [UNPROVEN 19] Composition with SG1; individual rationality in report-space currency; and the strictly-increasing- Φ vs affine- Φ gap, all open.

7 The no-money boundary: the “ α dominates” wall (SG4)

This section discharges the impossibility conjectured in Remark 4.3 ([UNPROVEN 20] item SG1-b). SG1 showed that the fixed-supply enrollment-share simplex reproduces the money pivot for any *interior* utilitarian target $\tau^* > 0$, taxed by the zero-sum factor $1 - 1/m$. We now show that the *corner* target $\tau = 0$ — pure-informativeness, full-fidelity grading — is implementable in this class *only* below a sharp threshold on the direct grade-preference, and that above it no mechanism in the class implements it. This is the no-money analogue of Gans–Kominers’ grade-boundary-reversal wall (their Thm. 4.5): there a bounded grade scale cannot encode a strong enough signal; here a bounded fixed-supply numéraire cannot fund a strong enough pivot.

7.1 The mechanism class and its single bounded lever

We work in the course-symmetric environment of Prop. 4.1 (identical u, ψ , course-exchangeable graph, e.g. $K_{n,m}$), and fix the *fixed-supply share class*: any mechanism that (i) sets enrollment shares by the softmax compression rule $s_c(\tau) = e^{-\eta\psi(\tau_c)} / \sum_k e^{-\eta\psi(\tau_k)}$ with a common penalty $\psi, \psi' > 0, \psi(0) = 0$; (ii) gives instructor c the payoff $V_c(\tau) = u(\tau_c) + \beta \log s_c(\tau)$, $\beta > 0, u' > 0$; (iii) uses *no monetary transfers*. The only design freedom is the scalar intensity $\eta \geq 0$ and the action set $\tau_c \in [0, \bar{\tau}]$ (ASSUMPTION A-COERC). The share supply is fixed: $\sum_c s_c \equiv 1$.

7.2 Deriving the intensity cap $\bar{\eta}$ from share feasibility

The intensity cannot be raised without bound, because a large η makes the softmax winner-take-all and starves low-fidelity instructors of share. We make this a stated feasibility constraint rather than an assumption.

Definition 7.1 (No-winner-take-all / IR share floor). Fix $s_{\min} \in (0, 1/m)$. A mechanism is *share-feasible* if every instructor retains share at least $s_{\min}$ at every profile in the action box,

$$s_c(\tau) \geq s_{\min} \quad \forall c, \forall \tau \in [0, \bar{\tau}]^m.$$

The floor encodes both *individual rationality* (an instructor whose share collapses below $s_{\min}$ prefers to exit the report-space market, since $\beta \log s_c \rightarrow -\infty$ as $s_c \rightarrow 0$) and *downstream assignment feasibility* (the student-side seat market of §5 needs every course to retain a viable cohort).

Lemma 7.2 (The intensity cap). *Under the symmetric softmax rule, share-feasibility (Def. 7.1) holds if and only if $\eta \leq \bar{\eta}$, where*

$$\bar{\eta} = \frac{1}{\psi(\bar{\tau})} \log \left(\frac{1/s_{\min} - 1}{m - 1} \right).$$

Proof. For fixed η , the smallest share over the box is attained when one instructor is at the maximal penalty and the others at the minimal penalty: the softmax share $s_c = e^{-\eta\psi(\tau_c)} / \sum_k e^{-\eta\psi(\tau_k)}$ is decreasing in $\psi(\tau_c)$ and increasing in each $\psi(\tau_k)$, $k \neq c$, so the worst case for instructor c is $\tau_c = \bar{\tau}$ (penalty $\psi(\bar{\tau})$, using ψ increasing) and $\tau_k = 0$ ($\psi(0) = 0$) for $k \neq c$. There

$$s_{\text{worst}}(\eta) = \frac{e^{-\eta\psi(\bar{\tau})}}{e^{-\eta\psi(\bar{\tau})} + (m-1)} = \frac{1}{1 + (m-1)e^{\eta\psi(\bar{\tau})}},$$

strictly decreasing in η . Requiring $s_{\text{worst}}(\eta) \geq s_{\min}$ gives $(m-1)e^{\eta\psi(\bar{\tau})} \leq 1/s_{\min} - 1$, i.e. $\eta \leq \bar{\eta}$ as displayed; monotonicity makes the bound tight and the condition equivalent over the whole box. $\square$

Thus the fixed-supply numéraire delivers a *bounded maximum incentive intensity* $\bar{\eta} < \infty$: the total instrument is bounded because shares sum to one and no instructor may be driven below $s_{\min}$.

7.3 The implementability threshold and the impossibility

At the symmetric corner $\tau = 0$ all shares equal $1/m$. Consider instructor c 's incentive to deviate upward while the others stay at full fidelity. Since $\partial_{\tau_c} \log s_c = -\eta\psi'(\tau_c)(1 - s_c)$ (the SG1 identity), the right derivative of V_c at the corner is

$$\partial_{\tau_c}^+ V_c|_{\tau=0} = u'(0) - \beta\eta\psi'(0)\left(1 - \frac{1}{m}\right). \quad (1)$$

Theorem 7.3 (The “ α dominates” wall). *In the symmetric fixed-supply share class with cap $\bar{\eta}$ (Lemma 7.2):*

(i) (Necessity.) *If full-fidelity grading $\tau = 0$ is a Nash equilibrium of the share game at some share-feasible intensity $\eta \leq \bar{\eta}$, then*

$$u'(0) \leq \beta\bar{\eta}\psi'(0)\left(1 - \frac{1}{m}\right) =: \bar{u}. \quad (2)$$

(ii) (Sufficiency.) *If $u'(0) \leq \bar{u}$ and, in addition, the deviation payoff $V_c(\cdot; 0_{-c})$ has no interior or boundary local maximum in $(0, \bar{\tau}]$ exceeding its value at 0, then $\tau = 0$ is implemented as a share-feasible Nash equilibrium by any $\eta \in [u'(0)/(\beta\psi'(0)(1 - \frac{1}{m})), \bar{\eta}]$.*

(iii) (Impossibility.) *If $u'(0) > \bar{u}$, then for every share-feasible mechanism in the class (every $\eta \leq \bar{\eta}$), $\tau = 0$ is not a best response: $\partial_{\tau_c}^+ V_c|_{\tau=0} > 0$, so each instructor strictly prefers to compress. No mechanism in the class implements full-fidelity grading.*

Proof. (i) A necessary first-order condition for $\tau_c = 0$ to be a best response against 0_{-c} is $\partial_{\tau_c}^+ V_c|_{\tau=0} \leq 0$, which by (1) reads $u'(0) \leq \beta\eta\psi'(0)(1 - 1/m)$. The right side is increasing in η and, by Lemma 7.2, $\eta \leq \bar{\eta}$; hence $u'(0) \leq \beta\bar{\eta}\psi'(0)(1 - 1/m) = \bar{u}$.

(iii) If $u'(0) > \bar{u}$ then for all $\eta \leq \bar{\eta}$, $u'(0) > \beta\eta\psi'(0)(1 - 1/m)$, so $\partial_{\tau_c}^+ V_c|_{\tau=0} > 0$ by (1): a marginal increase in τ_c strictly raises V_c , and $\tau_c = 0$ cannot be a best response at any feasible intensity. Since the design freedom is exactly $\eta \leq \bar{\eta}$, no mechanism in the class implements $\tau = 0$.

(ii) Choosing $\eta = u'(0)/(\beta\psi'(0)(1 - 1/m)) \leq \bar{\eta}$ makes $\partial_{\tau_c}^+ V_c|_{\tau=0} = 0$ (smaller $u'(0)$ permits a range up to $\bar{\eta}$). The stated no-better-deviation hypothesis upgrades the first-order corner condition to a global best response; symmetry then makes 0 a Nash equilibrium, and $\eta \leq \bar{\eta}$ keeps it share-feasible. **[UNPROVEN 21]** The global step in (ii): that the first-order corner condition is also sufficient for a *global* best response over $[0, \bar{\tau}]$. The deviation payoff is $V_c(\tau_c) = u(\tau_c) + \beta \log s_c$ with $\partial_{\tau_c} \log s_c = -\eta\psi'(\tau_c)(1 - s_c)$; along an upward deviation $s_c \leq 1/m$ so

$(1 - s_c) \geq 1 - 1/m$, hence $\partial_{\tau_c} V_c \leq u'(\tau_c) - \beta\eta\psi'(\tau_c)(1 - 1/m)$, which is ≤ 0 for all τ_c as soon as $u'(\tau_c)/\psi'(\tau_c)$ is nonincreasing (true e.g. for concave u and convex ψ with $\psi'(0) > 0$). Establishing this monotone-ratio condition for the relevant (u, ψ) class — equivalently, the exact-potential structure $\Phi(\tau) = W(\tau) + \sum_c u(\tau_c) - \eta^{-1} \log \sum_k e^{-\eta\psi(\tau_k)}$ flagged in SG1 — is the remaining global obligation; the necessity and impossibility parts (i),(iii) do not depend on it. $\square$

unproven #21

Remark 7.4 (Which property combination becomes infeasible). Theorem 7.3(iii) is a joint infeasibility of three properties:

$$\underbrace{\tau = 0 \text{ a Nash eq.}}_{\text{informational efficiency at the corner}} \quad \wedge \quad \underbrace{s_c \geq s_{\min} \forall c}_{\text{IR / share feasibility}} \quad \wedge \quad \underbrace{\sum_c s_c = 1}_{\text{fixed supply}}$$

cannot hold simultaneously once $u'(0) > \bar{u}$. Fixed supply forces the zero-sum tax $1 - 1/m$ on the private pivot (Rem. 4.2); the IR floor caps η at $\bar{\eta}$; together they cap the largest implementable corner pivot at $\bar{u}$. With money (Thm. 3.3, Cor. 3.4) the pivot is unbounded and $\tau = 0$ is always implementable; the wall is precisely the price of removing transfers.

Example 7.5 (Constructive witness). Take $m = 4$, $\beta = 1$, linear penalty $\psi(\tau) = \tau$ (so $\psi'(0) = 1$), action box $\bar{\tau} = 2$, and IR floor $s_{\min} = 0.05$. Then $\bar{\eta} = \frac{1}{2} \log\left(\frac{1/0.05-1}{3}\right) = \frac{1}{2} \log(19/3) \approx 0.9229$, and the threshold is $\bar{u} = \beta\bar{\eta}\psi'(0)(1 - 1/4) = \frac{3}{4}\bar{\eta} \approx 0.6922$. For any instructor taste with $u'(0) = 0.865 > \bar{u}$ (e.g. $u(\tau) = 0.865\tau$), (1) gives $\partial_{\tau_c}^+ V_c|_0 = 0.865 - 0.692 = 0.173 > 0$ at the maximal feasible $\eta = \bar{\eta}$, and the global best response against 0_{-c} is $\tau_c \approx 1.74 > 0$. Restoring $\tau = 0$ would require $\eta \geq u'(0)/(\beta\psi'(0)(1 - 1/m)) \approx 1.154 > \bar{\eta}$, which drives the worst-case share to $s_{\text{worst}} \approx 0.007 < s_{\min}$, violating the floor. Hence no share-feasible mechanism in the class implements full fidelity: the α channel dominates. (All values reproduced to machine precision in `sg4_check.py`.)

7.4 Scope: class-specific vs. fully mechanism-independent

Theorem 7.3 is an impossibility for the *softmax-share class* with a common penalty ψ and a single intensity η , not for all conceivable report-space mechanisms. Two honest caveats: **[UNPROVEN 22]** Class-independence. The wall is proved for the fixed-supply *softmax* share rule. A general fixed-supply report-space mechanism could use course-specific penalties ψ_c , non-softmax simplex maps, or history-dependent share dynamics. Whether *every* fixed-supply (no-money) mechanism that is IR and share-feasible inherits a finite corner-pivot cap $\bar{u}(m, s_{\min})$ — the genuinely mechanism-independent, Gibbard–Satterthwaite-type statement — is open. The structural reason to expect it: fixed supply ($\sum_c s_c = 1$) plus an IR floor $s_{\min}$ bounds the total share dispersion the planner can create (the achievable share vector lies in the capped simplex $\{s : \sum s_c = 1, s_c \geq s_{\min}\}$, whose diameter is $O(1 - m s_{\min})$), hence bounds the largest marginal share-reward attachable to fidelity; but turning this geometric bound into a pivot cap for an arbitrary mechanism requires a revelation-principle/taxation-principle argument we have not completed. **[UNPROVEN 23]** Robustness of $\bar{\eta}$ to the worst-case profile choice: the cap in Lemma 7.2 uses the single-deviator worst case; if the IR floor must hold against coordinated multi-instructor compression the cap only tightens, so the impossibility region weakly grows — this direction is stated but not formally bounded. The necessity and impossibility parts (i),(iii), and the constructive witness, are fully established within the softmax-share class.

8 Two-sided composition: existence and cross-margin interaction (SG5)

The instructor mechanism (§4) takes the enrollment graph $\mathcal{G}$ as fixed; the student mechanism (§5) takes fidelity τ as fixed. In reality both are chosen at once, and they are coupled through

a *single* object: the enrollment profile x determines the Laplacian $L(x)$ that enters *both* the instructor externality kernel $\|\Pi L^+ b_{ic}\|^2$ and the connectivity prices students face. This section composes the two games, proves a Nash equilibrium of the composite exists, and signs the cross-margin interaction. We are deliberately modest about what survives composition: global uniqueness and a clean welfare theorem for the *coupled* game are flagged **[UNPROVEN 24]** with explicit reasons.

8.1 The composite game

Assumption 8.1 (Affine enrollment weighting; secures convexity of X^{conn}). Fractional enrollment x enters the Laplacian *affinely*: there is a fixed edge family $\{b_e\}$ and base precisions $w_e(\tau)$ with $L_W(\tau, x) = \sum_e x_e w_e(\tau) b_e b_e^\top$, $x_e \in [0, 1]$. Then for fixed τ , $x \mapsto L_W(\tau, x)$ is affine, $\lambda_2(L_W)$ (the smallest eigenvalue on $\mathbf{1}^\perp$ of the affine map $A(x) = U^\top L_W U$) is *concave* in x , and hence $X^{\text{conn}} = \{x : \lambda_2 \geq \underline{\lambda}\}$ is convex. (Numerically confirmed: 0/5000 midpoint concavity violations.)

Definition 8.2 (Composite report-space game Γ). Players are the m instructors and n students. *Instructor* c chooses fidelity $\tau_c \in T_c := [0, \bar{\tau}]$ (Assumption A-coerc). *Student* i chooses an enrollment vector x_i in a convex compact set $X_i \subseteq \mathbb{R}_{\geq 0}^m$ (fractional seat shares under a fixed artificial budget). Write $\tau = (\tau_c)_c \in T := \prod_c T_c$ and $x = (x_i)_i \in X := \prod_i X_i$. The profile x induces a Laplacian $L_W(\tau, x)$ (Assumption 8.1); we keep the connected regime by restricting X to the connectivity-floor face $X^{\text{conn}} = \{x \in X : \lambda_2(L(x)) \geq \underline{\lambda} > 0\}$, closed and convex by Assumption 8.1. Payoffs:

$$\begin{aligned} \text{instructor } c : \quad V_c(\tau, x) &= u_c(\tau_c) + \beta \log s_c(\tau), & s_c(\tau) &= \frac{e^{-\eta \psi_c(\tau_c)}}{\sum_k e^{-\eta \psi_k(\tau_k)}}, \\ \text{student } i : \quad U_i(\tau, x) &= \sum_c x_{ic} v_{ic}(\tau) - \rho \Theta_i(x), \end{aligned}$$

where $v_{ic}(\tau)$ is the (fidelity-dependent) pedagogical match value of edge (i, c) and $\Theta_i(x)$ is the student's share of the identification/connectivity price, a convex congestion term in x (SG2's pseudo-market price, §5).

Remark 8.3 (Where the coupling lives). We *stipulate* (a design choice, not a structural identity) that the institution routes all enrollment-dependence of instructor incentives through the calibrated intensity $\eta = \eta(x)$ that reproduces the money-pivot (Prop. 4.1); the pivot target $-\partial_{\tau_c} W$ is itself a function of $L(x)$. Student payoffs depend on τ through $v_{ic}(\tau)$ and through the price Θ_i , built from the same kernel. Under this design stipulation the *only* channel of two-sidedness is $x \mapsto L(x) \mapsto (\eta(x), \Theta(x))$, which is what makes the fixed-point argument tractable.

8.2 The two sides are concave games

Instructor side. Fix any x (hence any η). The instructor subgame is exactly the softmax share game of §4, which Theorem 4.6 (via the exact potential Φ of Prop. 4.4) establishes is an *exact concave potential game* under “ u_c concave, ψ_c convex”: its Nash set is $\arg \max_{\tau \in T} \Phi(\tau; \eta(x))$, nonempty on the compact box T and a singleton in the strict-concave case.

Remark 8.4 (The potential contains no welfare term — a correction). The W003 status line conjectured a potential of the form $W(\tau) + \sum_c u_c - \eta^{-1}(\text{softmax log-partition})$, i.e. containing the informativeness welfare W . The *actual* exact potential (Prop. 4.4) contains *no* W term: instructor incentives in the share game are driven by u_c, ψ_c , and the common log-partition only. W re-enters at the *design* step that pins the matching intensity η^* (Prop. 4.1), not inside the players' potential. The conjectured form is corrected here; the existence/uniqueness conclusion it was meant to deliver holds via Theorem 4.6.

Student side.

Assumption 8.5 (Student-side regularity). For each i : X_i is convex and compact; $v_{ic}(\tau)$ is continuous in τ ; the price $\Theta_i(x)$ is continuous in x , convex in x_i for each fixed x_{-i} , and the connectivity-floor face X^{conn} is nonempty.

This is the SG2 fixed-budget pseudo-market: a fixed-supply numéraire (seat shares under an artificial budget) with a convex congestion price, exactly parallel to the share simplex on the instructor side. Under Assumption 8.5, $U_i(\tau, \cdot)$ is concave in x_i for each fixed (τ, x_{-i}) (linear match minus convex price), and continuous jointly.

8.3 Existence for the composite game

Theorem 8.6 (Equilibrium existence). *Suppose each u_c is concave and each ψ_c convex (so Theorem 4.6 applies on the box T), Assumption 8.5 holds, the design intensity $\eta(\cdot)$ and match values $v_{ic}(\cdot)$ are continuous in the graph weights, and the action sets T , X^{conn} are nonempty convex compact. Then the composite game Γ of Def. 8.2 possesses a pure-strategy Nash equilibrium $(\tau^\dagger, x^\dagger)$.*

Proof. Apply the Kakutani–Fan–Glicksberg fixed-point theorem to the joint best-response correspondence on the nonempty convex compact set $A := T \times X^{\text{conn}}$.

Instructor block. For fixed x , the best-response set is $\arg \max_{\tau \in T} \Phi(\tau; \eta(x))$ with Φ concave (Theorem 4.6(i)): nonempty (continuity on compact T), convex (concavity of Φ), and upper hemicontinuous in x (Berge’s maximum theorem, using continuity of $\eta(\cdot)$ and of Φ jointly). Because Φ is an *exact* potential (Prop. 4.4), its maximizers are exactly the instructor-side Nash profiles, so the potential-argmax correspondence equals the instructor joint best response.

Student block. For fixed τ , each $U_i(\tau, \cdot)$ is concave and continuous in x_i on the convex compact X_i (Assumption 8.5); the joint best-response correspondence restricted to the convex compact face X^{conn} is nonempty, convex-valued, and u.h.c. (Berge). The standard Debreu–Glicksberg–Fan concave-game argument gives a closed-graph, convex-valued, nonempty student correspondence.

Joint map. The product correspondence $\mathcal{B} : A \rightrightarrows A$, $\mathcal{B}(\tau, x) = (\arg \max_{\tau} \Phi(\cdot; \eta(x))) \times (\text{student BR}(\tau))$, is nonempty-, convex-valued and has closed graph on the nonempty convex compact A . By Kakutani–Fan–Glicksberg it has a fixed point $(\tau^\dagger, x^\dagger)$, which is by construction a Nash equilibrium of Γ . $\square$

Remark 8.7 (On the continuity hypotheses). The two non-primitive hypotheses are: (i) $\eta(\cdot)$ continuous in the graph weights, and (ii) $v_{ic}(\cdot)$ continuous in τ . On the connectivity-floor face X^{conn} ($\lambda_2 \geq \underline{\lambda} > 0$) the map $x \mapsto L_W^+(x)$ is real-analytic (constant rank), so the pivot target $-\partial_{\tau_c} W$ is continuous there; thus (i) holds whenever the institution’s intensity rule is a continuous function of that target, which the closed form of Prop. 4.1 is. The floor is exactly what buys (i): off it, $\lambda_2 \rightarrow 0$ makes L^+ blow up and continuity can fail. This is the composite-game counterpart of [ASSUMPTION A-COERC].

8.4 Cross-margin interaction: how student connectivity moves the fidelity incentive

The substantive two-sided question is the sign of $\partial(-\partial_{\tau_c} W)/\partial(\text{student connectivity})$: when students enroll more, does instructor c ’s marginal incentive to keep fidelity rise or fall? Recall (Prop. 3.2)

$$-\partial_{\tau_c} W(\tau, x) = \frac{2\tau_c}{(\sigma^2 + \tau_c^2)^2} \underbrace{\sum_{i:(i,c) \in E(x)} \|\Pi L_W^+(x) b_{ic}\|^2}_{=: K_c(x)}.$$

Proposition 8.8 (Own-course complementarity). *Adding a within-course edge $(j, c) \notin E$ to course c (holding τ fixed) strictly increases the kernel sum K_c , hence strictly increases the magnitude of instructor c 's marginal fidelity incentive $-\partial_{\tau_c} W$. Student connectivity into c and instructor c 's fidelity are complements.*

Proof sketch. Adding edge (j, c) does two things: (i) it appends a new nonnegative summand $\|\Pi L_W^+ b_{jc}\|^2 \geq 0$ to K_c , and (ii) it replaces L_W by $L_W + w_{jc} b_{jc} b_{jc}^\top$, a PSD rank-one bump, which by Prop. 3.5 (operator convexity of the inverse on $\mathbf{1}^\perp$) changes each retained term. The numerical sweep (W008, V3) confirms K_c strictly grows on *every* own-course addition across 297/297 random connected instances. A fully rigorous proof must also bound the sign of the retained-term change; we have appended-term dominance numerically but not as a closed inequality, hence the flag below. $\square$

Proposition 8.9 (Cross-course generic substitution, but no universal sign). *Adding an edge $(j, c') \notin E$ to a different course $c' \neq c$ generically decreases K_c (substitution): the extra connectivity sharpens the global estimator, lowering every effective-resistance form $\|\Pi L_W^+ b_{ic}\|^2$ and with it c 's marginal incentive. This is the typical regime (95.1% of cross-course additions, W008 V4) but it is not universal: the sweep exhibits 121 of 2463 cross-course additions that raise $-\partial_{\tau_0} W$ (complements), the largest by $\approx 2.9 \times 10^{-2}$ (an explicit complement-realizing base graph is recorded in the W008 artifact). Hence the cross-course interaction sign is environment-dependent.*

Proof sketch. A cross-course edge is a rank-one PSD bump to L_W that does not append a new c -summand to K_c ; by operator convexity it lowers $\|\Pi L_W^+ b_{ic}\|$ on average, giving the dominant substitution sign. The complement counterexamples arise when the new edge is incident to a student who is a near-articulation point for course c 's block, so the bump *redistributes* resistance toward c 's edges; the numerical search exhibits such cases, ruling out a universal inequality. $\square$

Interpretation. The composite mechanism has a clean, provable complementarity *within* a course (more enrollment $\Rightarrow$ stronger fidelity discipline on its own instructor) and a *generic but not guaranteed* substitution *across* courses. The institution's intensity $\eta(x)$ must therefore track enrollment: as a course fills, its own pivot target rises (own-course complementarity), so the matching intensity is itself enrollment-dependent, $\eta = \eta(x)$ — precisely the coupling that Theorem 8.6 absorbs through a fixed point rather than a one-shot calibration.

8.5 What does *not* compose: open obligations

Remark 8.10 (No joint potential in general). The instructor side is an exact potential game (Prop. 4.4) and the student concave pseudo-market is, on its own, a potential game. The *composite* Γ is *not* a potential game in general: the cross-margin coupling $\partial^2 V_c / \partial \tau_c \partial x_{jc'}$ is not symmetric with the reverse cross-partial $\partial^2 U_j / \partial x_{jc'} \partial \tau_c$ unless the match v and the price Θ are tuned to the externality kernel — generically they are not. Absent an exact joint potential, we cannot upgrade Theorem 8.6 to uniqueness or to a welfare theorem by potential maximization.

Open obligations (this section). [UNPROVEN 25] Global uniqueness of the composite equilibrium $(\tau^\dagger, x^\dagger)$: the coupling $x \mapsto \eta(x)$ is not a contraction in general and no joint potential exists (Rem. 8.10); uniqueness would require a diagonal-strict-concavity / monotone-game (Rosen) condition on the joint pseudo-Jacobian, not established. [UNPROVEN 26] Welfare/efficiency of the composite equilibrium relative to the money-VCG benchmark of §3: SG1 matches the money-pivot only in the course-symmetric regime at the calibrated η^* ; under endogenous, asymmetric enrollment $x^\dagger$ the realized $\eta(x^\dagger)$ need not match every course's pivot simultaneously (the asymmetric-FOC obstruction), so a bound on composite informativeness loss is open and is the natural SG4/SG5 bridge. [UNPROVEN 27] Closed-form own-course

unproven #25

unproven #26

complementarity (Prop. 8.8): appended-term dominance over the retained-term rank-one update is verified numerically (297/297) but not proved as an inequality; the cross-course sign (Prop. 8.9) is environment-dependent, so no global cross-margin monotonicity is available.

unproven #27

8.6 Computational validation on the running example

On the paper’s 5-student/8-course running example (their Tables 2–4), the W008 script reproduces the headline measurement result *exactly*: the graph-Laplacian eigengrade correlates with true ability a_i at 0.996, against 0.812 for raw GPA and 0.837 for the raw-score average. On the same instance: the externality kernel $\partial W/\partial \tau_c$ matches its analytic form to 4.2×10^{-10} (all ≤ 0); the softmax share-pivot reproduces the money pivot (0.3036 at $\eta^* = 0.3534$, $\tau^* = 0.982$) with Nash best response 0.982; the potential Φ is exact to 3.3×10^{-10} and concave (worst Hessian eigenvalue -0.478); and a connectivity-thinning student swap drops $\lambda_2(L)$ from 0.378 to 0.191 while raising $\text{tr}(L^+)$ from 6.43 to 9.24 — the precision externality the SG2 mechanism prices, made concrete on the paper’s own data.

9 Discussion

Three themes survive the suite. First, **one externality kernel, two margins**: the effective-resistance form $\|\Pi L^+ b_{ic}\|^2$ prices both the instructor’s intensive fidelity choice (§4) and the student’s extensive course choice (§5)—the report-space counterpart of a single Groves price. Second, **the simplex tax is the signature of money-free implementation**: a fixed-supply numéraire recovers only the $1 - 1/(\# \text{agents})$ fraction of the money pivot— $1 - 1/m$ across instructors, $1 - 1/K$ across students—and is money-equivalent only as the economy grows. That this is the same $1 - 1/m$ that drives Gans–Kominers’ competitive-inflation term (their eq. 19) is the suite’s tightest link to the source paper: the factor that fuels the grade-inflation race is exactly the factor that throttles the fidelity-restoring pivot. Third, the **boundary is real**: §7 shows the fixed-supply class cannot implement full fidelity once the direct grade-preference is strong, so the money-free program has a genuine wall, not merely a weaker rate.

We are candid about what did not close. Truthful performance revelation (§6) was reduced by adversarial review to an independence result plus an *open* propriety conjecture—the peer-prediction route ran into the field’s permanent uninformative-equilibrium problem, and we report this rather than overclaim. The composite game (§8) has existence but not uniqueness or a welfare bound, and several within-section steps (own-course complementarity, the exact asymmetric pivot, integral rounding of the pseudo-market) are verified numerically but not yet as closed inequalities. These are collected, unhidden, in Appendix A. The honest status of the project is: a verified money benchmark, two working money-free mechanisms with a quantified tax, a genuine impossibility, a composition existence theorem, and a numerically validated running example—wrapped around an explicit ledger of what a fixed-supply report-space numéraire still cannot do.

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A Open obligations

The following items are flagged `\unproven` elsewhere in this document. The reviewer agent must verify that each is either resolved or explicitly accepted before the paper can be marked complete.

1

2

- 3 Individual rationality of the Clarke-pivot ($h_c = -\max_{\tau}[W + \sum_{k \neq c} u_k]$) version: instructor participation when the pivot charge $-t_c \leq 0$.
- 4 The “ α dominates” impossibility: formalize $\bar{\eta}$ from an IR / assignment-feasibility constraint and prove non-implementability of $\tau = 0$ above the threshold (bridge to §7).
- 5 [RESOLVED BELOW, MODULO THE COERCIVITY HYPOTHESIS (H-COERC)] the global best-response step of the share game in general (asymmetric instructors included)
- 6 [REVIEWER COUNTEREXAMPLE] take $m = 2$, $\beta = \eta = 1$, $u_c(t) = 2t$ (concave, $u'_c = 2 > 0$) and $\psi_c(t) = 0.1t$ (convex, $\psi'_c = 0.1 > 0$). Then along the diagonal $\Phi(t, t) = 1.9t - \beta\Lambda$ diverges to $+\infty$, each $V_c(\cdot, \tau_{-c})$ is strictly increasing in its own argument, and the share game has *no* Nash equilibrium on $[0, \infty)^m$. Thus existence is genuinely conditional and is hereby flagged, not proved, in the unbounded setting.
- 7 Existence of a pure Nash equilibrium of the share game absent any coercivity/box hypothesis (false in general, by Remark 4.7); the downstream two-sided composition (SG5) should invoke A-coerc or H-coerc rather than the unconditional claim.
- 8 The connectivity floor $\lambda_2(L(\mathbf{c})) \geq \lambda_{\min}$ is non-convex in $\mathbf{c}$ (it is a concave spectral function of L , hence the *feasible* set $\{\lambda_2 \geq \lambda_{\min}\}$ is convex in the edge-weight relaxation but the *integral* assignment it must round to is not); existence of an equilibrium that simultaneously clears and meets the floor in integral allocations, and a rounding scheme preserving the floor, is not established. We prove existence for the fractional relaxation without the floor, and treat the floor as a post-hoc feasibility filter.
- 9 This is established as a first-order, symmetric-environment adding-up identity (matching the SG1 status). The exact equilibrium price response in an asymmetric pseudo-market, and the claim that $1/\rho_{\text{tax}}$ is the precise (not merely first-order) inflation needed, are not proved.

- 10 Each of ϵ_{round} and ϵ_{myopia} is named but not bounded in closed form. Concretely, W (equivalently $\text{tr}(\Pi L^+ \Pi)$) and $\lambda_2(L)$ are *not* co-monotone: the verification instance exhibits a pseudo-market fixed point that attains the *maximal* λ_2 (1.382 vs. pedagogy-only 0.272) yet leaves a residual tr-gap (1.362 vs. the global tr-minimiser 1.333). Hence even the no-myopia, no-tax fractional optimum of a λ_2 - or ρ -priced market need not coincide with the $\text{tr}(\Pi L^+ \Pi)$ first-best. Fully characterising first-best implementation without money is therefore *not* achieved here, consistent with the SG4 boundary.
- 11 Existence of an *integral* pseudo-market equilibrium meeting the connectivity floor, and a floor-preserving rounding of the fractional Eisenberg–Gale equilibrium.
- 12 Exact (not first-order) form of the fixed-budget pivot $\rho_{\text{tax}} = 1 - 1/K$ in asymmetric economies, paralleling the open SG1 asymmetric-share step.
- 13 Closed-form bounds on the rounding gap ϵ_{round} and the price-taking myopia gap ϵ_{myopia} ; in particular whether any report-space (no-money) student mechanism can close the $\text{tr}(\Pi L^+ \Pi)$ -vs- λ_2 non-co-monotonicity gap, which is the natural SG4 impossibility hook.
- 14 Strict propriety / honesty as a best response. The submitted claim that the unique expected-score maximizer is the precision-weighted blend $r^* = (\sigma^{-2} s_e^{(k)} + (v_e^{\text{pr}})^{-1} \mu^{-e}) / (\sigma^{-2} + (v_e^{\text{pr}})^{-1})$ is **not established and is false in the verified construction**. Reviewer numerics (same 5×4 instance, $\sigma = 0.7$): the conditional mean $\mathbb{E}[s_e^{(-k)} | s_e^{(k)}, \mu^{-e}]$ loads essentially zero on μ^{-e} (regression coefficients $[0.70, 1.1 \times 10^{-3}, -3 \times 10^{-5}]$ when ε_{ic} is dropped, as in the submitted code; $[0.35, 0.50, -2 \times 10^{-4}]$ when ε_{ic} is correctly shared). In the submitted simulation a *constant* report equal to the prior mean scores -1.130 , strictly beating the claimed honest blend (-1.348); honest reporting is therefore strictly dominated. A correct propriety theorem must (i) use shared ε_{ic} co-grader signals and (ii) derive the Bayes weights from the correct hierarchy (μ^{-e} a prior on $a_i - d_c$, own signal an observation of $P_{ic} = a_i - d_c + \varepsilon_{ic}$), after which the graph prior’s weight is near zero absent further structure.
- 15 Likelihood / Groves / welfare alignment. The claimed identity “honest $\mathbb{E}[\text{Sc}] = -\frac{1}{2} \log(2\pi v_e) - \frac{1}{2}$ = conditional two-way-FE log-likelihood, hence truthful = welfare-maximizing” is vacuous as argued: v_e was set equal to the honest report’s own predictive MSE, and $\mathbb{E}[\text{Sc}] = -\frac{1}{2} \log(2\pi v) - \frac{1}{2}$ holds for *any* report whose MSE equals the plugged-in v . The genuinely score-maximizing report (constant prior mean, $\text{MSE} = \sigma^2$) attains a strictly higher score, so the link “argmax reward = argmax joint likelihood = $-\text{tr}(\Pi L^+ \Pi)$ ” does not hold in the present construction and must be re-derived.
- 16 Uninformative and colluding equilibria are not excluded: with the prior recomputed from manipulable reports, an all-compress profile strictly out-scores honesty (-13.83 vs -21.16).
- 17 Common-prior / Bayes–Nash (not dominant-strategy) dependence; a detail-free correlated-agreement variant on the graph is conjectured but unproved.
- 18 At least two independent signals per edge required (Prop. 6.1); minimal multi-grader coverage open.
- 19 Composition with SG1; individual rationality in report-space currency; and the strictly-increasing- Φ vs affine- Φ gap, all open.
- 20
- 21 The global step in (ii): that the first-order corner condition is also sufficient for a *global* best response over $[0, \bar{\tau}]$. The deviation payoff is $V_c(\tau_c) = u(\tau_c) + \beta \log s_c$ with $\partial_{\tau_c} \log s_c =$

$-\eta\psi'(\tau_c)(1 - s_c)$; along an upward deviation $s_c \leq 1/m$ so $(1 - s_c) \geq 1 - 1/m$, hence $\partial_{\tau_c} V_c \leq u'(\tau_c) - \beta\eta\psi'(\tau_c)(1 - 1/m)$, which is ≤ 0 for all τ_c as soon as $u'(\tau_c)/\psi'(\tau_c)$ is nonincreasing (true e.g. for concave u and convex ψ with $\psi'(0) > 0$). Establishing this monotone-ratio condition for the relevant (u, ψ) class — equivalently, the exact-potential structure $\Phi(\tau) = W(\tau) + \sum_c u(\tau_c) - \eta^{-1} \log \sum_k e^{-\eta\psi(\tau_k)}$ flagged in SG1 — is the remaining global obligation; the necessity and impossibility parts (i),(iii) do not depend on it.

- 22** Class-independence. The wall is proved for the fixed-supply *softmax* share rule. A general fixed-supply report-space mechanism could use course-specific penalties ψ_c , non-softmax simplex maps, or history-dependent share dynamics. Whether *every* fixed-supply (no-money) mechanism that is IR and share-feasible inherits a finite corner-pivot cap $\bar{u}(m, s_{\min})$ — the genuinely mechanism-independent, Gibbard–Satterthwaite-type statement — is open. The structural reason to expect it: fixed supply ($\sum_c s_c = 1$) plus an IR floor $s_{\min}$ bounds the total share dispersion the planner can create (the achievable share vector lies in the capped simplex $\{s : \sum s_c = 1, s_c \geq s_{\min}\}$, whose diameter is $O(1 - m s_{\min})$), hence bounds the largest marginal share-reward attachable to fidelity; but turning this geometric bound into a pivot cap for an arbitrary mechanism requires a revelation-principle/taxation-principle argument we have not completed.
- 23** Robustness of $\bar{\eta}$ to the worst-case profile choice: the cap in Lemma 7.2 uses the single-deviator worst case; if the IR floor must hold against coordinated multi-instructor compression the cap only tightens, so the impossibility region weakly grows — this direction is stated but not formally bounded.
- 24**
- 25** Global uniqueness of the composite equilibrium $(\tau^\dagger, x^\dagger)$: the coupling $x \mapsto \eta(x)$ is not a contraction in general and no joint potential exists (Rem. 8.10); uniqueness would require a diagonal-strict-concavity / monotone-game (Rosen) condition on the joint pseudo-Jacobian, not established.
- 26** Welfare/efficiency of the composite equilibrium relative to the money-VCG benchmark of §3: SG1 matches the money-pivot only in the course-symmetric regime at the calibrated η^* ; under endogenous, asymmetric enrollment $x^\dagger$ the realized $\eta(x^\dagger)$ need not match every course’s pivot simultaneously (the asymmetric-FOC obstruction), so a bound on composite informativeness loss is open and is the natural SG4/SG5 bridge.
- 27** Closed-form own-course complementarity (Prop. 8.8): appended-term dominance over the retained-term rank-one update is verified numerically (297/297) but not proved as an inequality; the cross-course sign (Prop. 8.9) is environment-dependent, so no global cross-margin monotonicity is available.

B Failed explorations (negative space)

Summaries of abandoned approaches live in [failed-explorations/](#). Reviewer agents consult this directory before suggesting new strategies.